

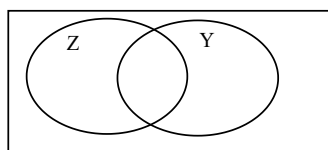
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Course: Calculus Beta

Assignment: MyLab Math - Uge 7

Let $P(Z) = 0.46$, $P(Y) = 0.36$, and $P(Z \cup Y) = 0.66$. Use a Venn diagram to find $P(Z' \cap Y')$, $P(Z' \cup Y')$, $P(Z' \cup Y)$, and $P(Z \cap Y')$.

Use the following Venn diagram showing sets Z and Y .



Determine the probability for each region of the diagram. Since you know $P(Z)$, $P(Y)$, and $P(Z \cup Y)$, you can use the union rule to calculate $P(Z \cap Y)$.

$$P(Z \cup Y) = P(Z) + P(Y) - P(Z \cap Y)$$

$$0.66 = 0.46 + 0.36 - P(Z \cap Y)$$

$$P(Z \cap Y) = 0.46 + 0.36 - 0.66$$

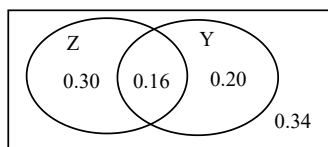
$$= 0.16$$

Since $P(Z) = 0.46$, then the region of Z that does not overlap Y is $0.46 - 0.16 = 0.30$.

Similarly, the region of Y that does not overlap Z is $0.36 - 0.16 = 0.20$.

The entire region of the Venn diagram must have probability 1. Since $P(Z \cup Y) = 0.66$, then the probability for the remaining region is $1 - 0.66 = 0.34$.

The completed Venn diagram looks like the following.



Use the diagram to find $P(Z' \cap Y')$.

Look at Z' , which includes the regions labeled 0.20 and 0.34. Then look at Y' , which includes the regions 0.30 and 0.34. $Z' \cap Y'$ is the region they have in common.

$$P(Z' \cap Y') = 0.34$$

Find $P(Z' \cup Y')$. The event $Z' \cup Y'$ includes the regions that are not in Z or are not in Y .

Add.

$$P(Z' \cup Y') = 0.30 + 0.20 + 0.34$$

$$= 0.84$$

Find $P(Z' \cup Y)$. The event $Z' \cup Y$ includes the regions labeled 0.16, 0.20, and 0.34.

Add.

$$P(Z' \cup Y) = 0.16 + 0.20 + 0.34$$

$$= 0.70$$

Find $P(Z \cap Y')$. The event $Z \cap Y'$ is the part of Z that does not overlap Y .

$$P(Z \cap Y') = 0.30$$